

DDTA research work plan after documentation-layer closure

WORK PLAN - REVISION 5

Status: active after BA0-T3 closure.

Supersedes: Revision 4 only for forward execution state; R1-R4 remain historical research records.

1. Fixed prior state

- Chapters 2-4: CLOSED / FINAL for current thesis scope.
- Documentation layer: CLOSED.
- BA0-R systems-modeling prior-art research: CLOSED.
- BA0-T1: COMPLETED / PROVISIONAL.
- BA0-T2: COMPLETED / PROVISIONAL PASS.
- BA0-T3: CLOSED.
- BA0 responsibility and non-goals: CLOSED.
- No BAE type or relation was accepted by BA0.
- ThreatForge remains a case study/tooling instrument, not DDTA semantic authority.

2. Closed Base Analysis responsibility

The authoritative BA0 responsibility boundary for forward methodology work is Base Analysis responsibility boundary R1.

Base Analysis is the accepted methodology-neutral analytical representation of shared project meaning required by DDTA. It preserves source/origin provenance, baseline-scoped semantic identity and explicit unresolved state sufficiently to support shared human/method projections, source drill-down, change-impact reasoning and source-localized feedback handoff.

Governed documentation remains project authority. Method-specific interpretation, analysis results, corrective policy decisions and documentation governance remain outside the shared Base Analysis core.

Semantic canonicity does not require one permanently materialized graph.

3. Closed BA0 responsibility set

Forward work must preserve:

1. BA0-C1 Authority and provenance boundary.
2. BA0-C2 Baseline-scoped shared semantic identity.
3. BA0-C3 Method-neutral shared core.
4. BA0-C4 Explicit uncertainty and diagnostic localization.
5. BA0-C5 Projection readiness and source drill-down.
6. BA0-C6 Change-impact traceability.
7. BA0-C7 Source-localized feedback handoff.
8. BA0-C8 Minimality and representation independence.

grounded, derived and diagnostic/unresolved are required responsibility-level distinctions. A generic reviewed analytical addition class is not required by BA0 and must not be assumed by BA1.

4. Sequence after BA0 closure

Do not skip phases:

BA1 - Minimal BAE ontology

BA2 - Relations and canonical action vocabulary

BA3 - Document-to-BAE derivation, provenance and authority

BA4 - Multi-level human and method projections

BA5 - Controlled vocabulary and optional authoring/extraction assistance

BA6 - Base Analysis closure and regression

BA0 closure authorizes ontology work; it does not pre-approve any type.

5. BA1 - Minimal BAE ontology

BA1-T1 - minimal BAE ontology candidate derivation

NEXT / NOT EXECUTED BY THIS PLAN REVISION.

Start from BA0-C1..C8, BA0-R prior-art pressure and governed DDTA corpora. For each recurring semantic responsibility required by the closed boundary, test whether it needs:

- an independently identified entity;
- a relation;
- a role;
- a property/attribute;
- a constraint/state;
- a provenance/diagnostic record;
- or only a derived/projection construct.

The first pass must prefer the smallest representation that preserves the closed responsibilities. **semantic responsibility exists** does not imply **first-class metaclass**.

Historical **actor/component/asset/boundary/data_flow** categories, STRIDE DFD constructs, ThreatForge implementation classes and SysML/KerML/ArchiMate/AADL concepts are candidate comparisons only. None is accepted by inheritance.

BA1-T1 exit

Produce a falsifiable minimal ontology candidate with explicit evidence for every proposed first-class identity and an explicit rejected/deferred list. Do not begin BA2 relation-vocabulary closure until the BA1 candidate has been pressure-tested.

6. BA2 - Relations and canonical action vocabulary

After BA1, define the smallest stable semantic relation vocabulary. Keep semantic relation, canonical lexical label and authoring synonym assistance separate.

7. BA3 - Document-to-BAE derivation, provenance and authority

Close exact identity/equivalence/lifecycle rules, source locators, grounded/derived/diagnostic material states, review boundaries, stale state and document-to-analysis traceability.

If a methodology-neutral shared-core analytical addition becomes demonstrably necessary here, do not silently introduce it. Reassess whether BA0 must reopen.

Automatic candidate extraction may be evaluated only as optional assistance and is not required for thesis validity.

8. BA4 - Multi-level human and method projections

Define projections over the same accepted Base Analysis semantics. Test progressive human navigation and method-specific views without creating a second authored project model. No view is source authority.

9. BA5 - Controlled vocabulary and optional assistance

Vocabulary reuse, duplicate suggestions, NLP/LLM proposals and candidate extraction are optional support. None may silently establish identity, equivalence, ownership, project fact or BAE acceptance.

10. BA6 - Base Analysis closure and regression

Regress closed corpora and at least one structurally different holdout. Verify semantic identity stability, source traceability, projection consistency, diagnostic behavior, change/re-analysis scope, method neutrality and human drill-down support.

BA6 closes the implemented Base Analysis design; BA0 has already closed only the responsibility boundary.

11. Analysis envelope after Base Analysis closure

Only after BA6:

- A1 - AnalysisRecord / execution envelope;
- A2 - common reviewed Finding boundary;
- A3 - change/provenance integration.

Then define the generic overlay contract and full methodology evaluations.

12. Method-specific evaluations

Planned concrete evaluations remain STRIDE and STRIDE-AI over the same common core. They are demonstrators of the overlay boundary, not proof of universal method support.

The BA0-T2 STRIDE probe remains only a pre-overlay consumer probe and does not count as formal STRIDE evaluation.

13. Corrective feedback boundary

The general DDTA loop remains:

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analysis / validation / diagnostic
-> source-localized corrective candidate
-> governed review
-> correction / clarification / supersession / new requirement or decision
-> Base Analysis revalidation
-> impacted re-analysis
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BA0-C7 requires traceable handoff to the governed source area. It does not assign the review/acceptance workflow to Base Analysis.

The thesis may still evaluate the narrower accepted security-Finding-to-SecurityRequirement path as a specific research contribution.

14. Next authorized microstep

Execute only:

BA1-T1 - minimal BAE ontology candidate derivation from the closed BA0 responsibilities.

Do not start BA2, formal STRIDE overlay design, Common Finding schema or implementation work during this microstep.